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Roll No .....

**EE/EX-7005(3)-CBGS**

**B.E. VII Semester**

Examination, December 2020

**Choice Based Grading System (CBGS)**

**FACTS**

**Time : Three Hours**

**Maximum Marks : 70**

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. a) What are the possible benefits of FACTS controllers?  
Discuss the classifications of FACTS controllers on basis of connection diagram.  
b) Enlist any ten examples of FACTS controllers. Also mention its practical installation in all over world (any five).
2. a) Discuss the mathematical formulations of SVC in load flow analysis and stability studies.  
b) What do you mean by “Power system” oscillation? How it is minimised by SVC?
3. a) Explain working and impedance VS delay angle characteristics of TCSC.  
b) Draw and explain GTO controlled series capacitor (GCSC)
4. a) Draw and explain V-I and V-Q characteristics of STATCOM.  
b) Explain control scheme and operation of STAT COM and SSSC.

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5.
  - a) Explain working principle and operation of UPFC.
  - b) What is FACTS controller interactions?
6.
  - a) What do you mean by optimal placement and property coordinated control of FACTS controller in power system environments? Explain in brief.
  - b) Discuss the co-ordination of multiple controllers using linear control techniques.
7.
  - a) What are factors which limits the loading capability?
  - b) Discuss the capabilities of Shunt compensation in improving the transient stability.
8. Describe the following. (any two)
  - a) Need for FACT controller
  - b) SSR mitigation
  - c) SVC-SVC interaction
  - d) Quantitative treatment of control co-ordination

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